6. A customer tries to vend without enough money

The customer presses the vend button before depositing the full price of the selected item.

Objective	• Prevent an item from being vended when the customer has not deposited enough money.	
Results	• The controller blocks the purchase and keeps the deposited money credited so the customer can add more money or cancel.	
Actors	• Customer, who understands the basic keypad, switches and buttons; vending machine controller.	
Related use cases	• Use Case 1: Buying an item. Use Case 3: Canceling a purchase. Use Case 5: Overpayment.	
Trigger	• The customer presses KEY3 while the deposited total is less than the selected item's price.	
Stakeholders	• The owner who needs to prevent unpaid purchases and preserve accurate customer balances.	
Resources	• Programmed SoC Series V board, and Quartus.	
Assumptions	• A working setup with the seven-segment display, working money switches, working RAM set up for Items and the working keypad as well.	
Post-condition	• No item has been vended. The selection and deposited balance remain available so the customer can continue or cancel.	
Frequency	• Whenever a customer attempts to purchase an item before paying its full price.	
Mission Priority	• High. The controller must prevent purchases with insufficient payment.	
Safety Priority	• Should hold no safety concern	
Flow of events	**Step #**	**Action**
	1.	The customer enters location 11 using the keypad.
	2.	The controller displays the selected items price at $1.35.
	3.	The customer moves SW3 on to simulate inserting $1.00. The controller displays a payment total of $1.00.
	4.	The customer presses KEY3.

	5.	The controller compares the $1.00 payment with the $1.35 price and blocks the vend.
	6.	The controller indicates that payment is insufficient and keeps the $1.00 credited to the customer.
Alternate Events	**Step #**	**Action**
	4A	The customer presses KEY3 with no money deposited. The controller blocks the vend and keeps the balance at $0.00.
	6A	The customer inserts a quarter using SW2 and a dime using SW1. The total becomes $1.35. After another KEY3 press, the controller simulates vending the item and displays zero change.
	6B	The customer presses KEY1 to cancel. The controller displays a full refund of $1.00 and clears the transaction.
Hazard/ Result /Mitigation	• If case is used incorrectly, the entire system is likely incorrect or not working to specifications.	
Requirements from Use Case	• Block vending whenever the deposited total is below the selected item's price. • Preserve the selection and deposited balance after an insufficient payment attempt. • Provide a visible indication that more payment is needed • Allow the customer to add money and try again. • Allow KEY1 to cancel and return the full deposited amount before a successful vend.	